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List It

Design and Implementation of Mobile Applications

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1. The Idea
2. Architecture
3. DEMO
4. Layouts
5. Testing



The Idea

- **Problem:** Grocery shopping is naturally inefficient.

1. Shopping list drafting.
2. Spending habit management.

- **Goal:**

- ▶ Automatic product categorization.
- ▶ Accounting system.



Architecture

Leonardo Ratti, Mattia Peruzzi

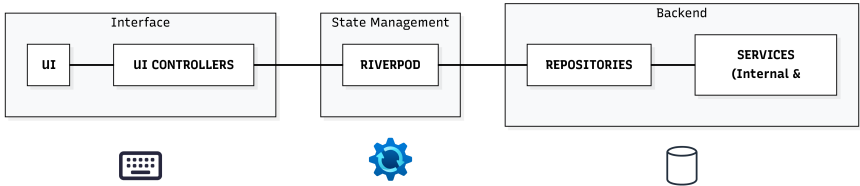
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The app is cross-platform. It supports:

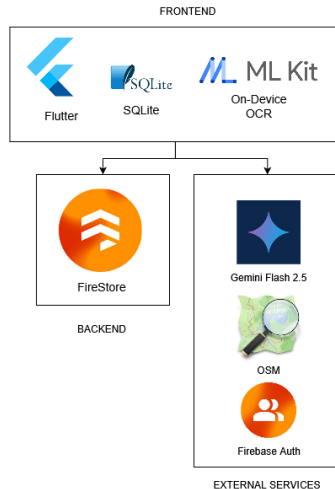
- **Android:** Android 5+
- **iOS:** iOS 13+, iPadOS 13+

Architecture Overview

4/23



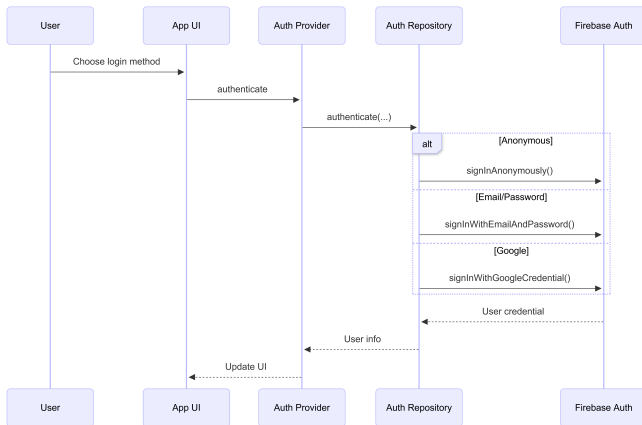
-  **UI:** Flutter
-  **Data:** SQLite & Firestore - User data, shopping lists, supermarkets, products, purchased products, categories
-  **API:** Firebase APIs, Overpass API, Gemini API
-  **Auth:** Firebase Auth & Google Sign-In
-  **LLM:** Google Gemini & Google ML Kit - Product categorization, recipe addition, data extraction.
-  **Maps:** OpenStreetMap - Nearest Supermarket

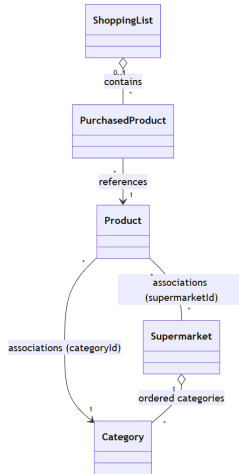




DEMO

- Anonymous User
- Email Sign In
- Google SSO
- Sync Engine

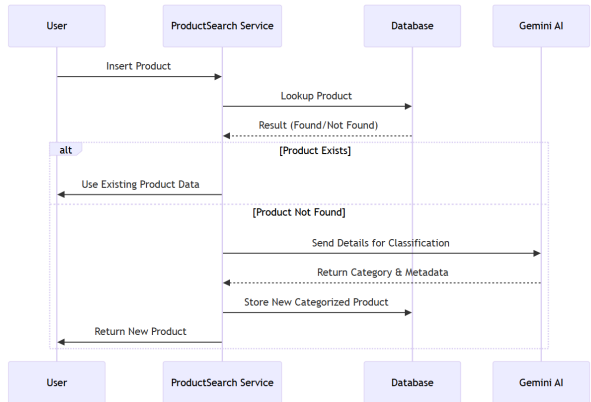




- **BottomNavigationRail** for fast access to core features: *Lists, History, Supermarket, statistics*
- **Navigator**
- **Side menu** for secondary features: *Profiles, Settings, Trash*

- **List Tab**
- **History Tab**
- **Supermarket Tab**
- **Statistics Tab**

- **Logic:** Initial database lookup with eventual **Gemini AI** classification if not already present in the database.
- **Outcome:** Existing product or new categorized product.

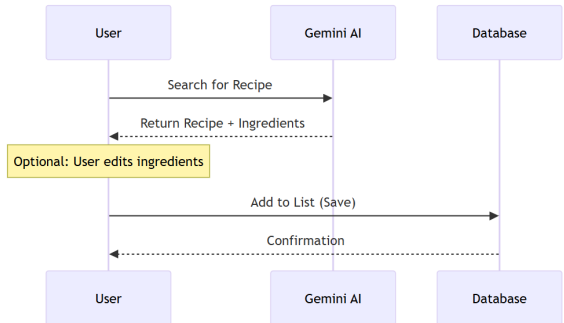


DEMO

Recipe Addition

11/23

- **AI Search:**
Asynchronous request to **Gemini**.
- **User Feedback:** User can decide if add or not each ingredient and modify each name.
- **Persistence:** Final data is stored asynchronously in the **Database**.



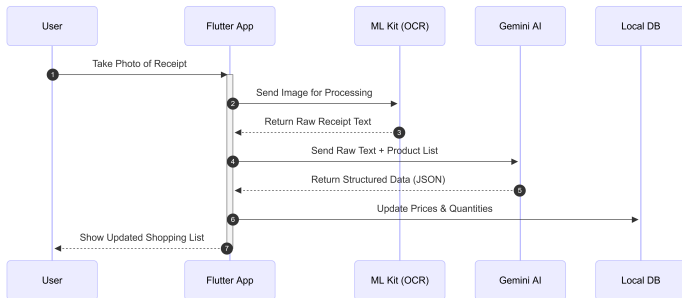
DEMO

Receipt Data Extraction

12/23

Three steps:

1. **Extraction**
2. **Mapping and Matching**
3. **Update**

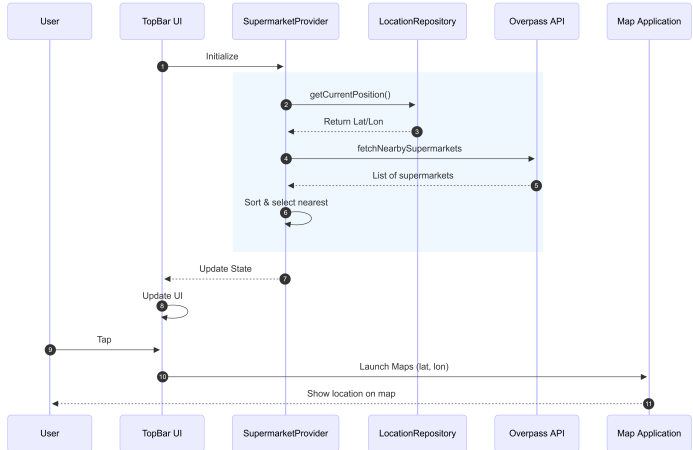


DEMO

Nearest Supermarket

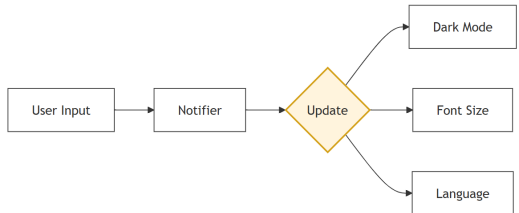
13/23

- Geolocator
- Overpass API
- OSM



- Per-list actions: **Restore** or **Delete Permanently**
- Bulk actions in header:
 - ▶ **Restore All**
 - ▶ **Empty Trash**

- **Dark Mode:** Toggle switch to enable or disable the global dark theme.
- **Font Size:** Dynamic slider (80%–140%).
- **Language:** Dropdown selector to switch between English and Italian.





Layout

- **Navigation Rail**
- **Dual-Pane Layout:** Detail Pane Navigator
- **Navigation Enhancement**



Testing

- **Libraries:** flutter_test, mocktail.
- **Mocking Environment**
- **Bottom Up Approach:** Unit - Widget - Integration
- **User Tests**

Coverage by Folder

Folder	Covered Lines	Total Lines	Coverage
lib/l10n	242	731	33.11% 
lib/models	452	459	98.47% 
lib/providers	619	813	76.14% 
lib/repositories	1040	1412	73.65% 
lib/screens	2722	3886	70.05% 
lib/services	1101	1240	88.79% 
lib/utils	262	290	90.34% 
lib/widgets	974	1222	79.71% 



Thank You



Appendix

- **Number of screens:** 19
- **Device Service:** Camera, GPS.
- **External Services:** Firebase, OSM, Gemini Flash.
- **Layout:** Adaptive mobile - tablet layout.
- **Look & Feel:** Simple and Intuitive.
- **Navigation:** Navigation bar, navigator object, side menu.

Category	Key Libraries
 Architecture	flutter_riverpod, provider, intl
 Data & Cloud	sqflite, firebase_core, cloud_firestore, firebase_auth
 AI & ML	google_generative_ai, google_mlkit_text_recognition
 Device Features	camera, geolocator, connectivity_plus, permission_handler
 Utilities	logger, uuid, flutter_staggered_grid_view
 Testing	flutter_test, mocktail, integration_test

/lib/

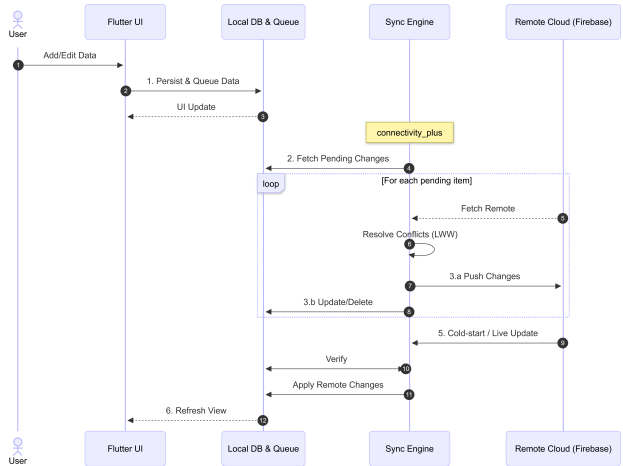
- **l10n/** - Multi language support.
- **models/** - Core entities.
- **providers/** - State management classes.
- **repositories/** - Abstractions for data access.
- **screens/** - UI views.
- **services/** - API clients.
- **styles/** - Theme, color, and style definitions.
- **utils/** - Utility functions and helper classes.
- **widgets/** Reusable UI components and custom widgets.
- **main.dart**

Appendix

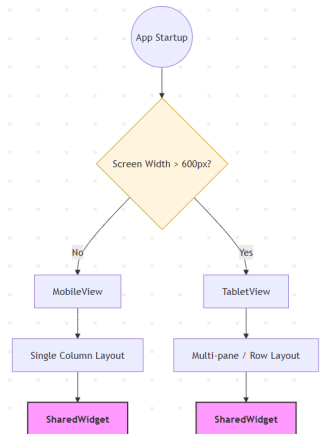
Sync Mechanism

21/23

- **Offline First**
- **Last Write Wins**
- **Silent Updates**
- **Soft Deletes**



- Device type detected at startup (phone vs tablet)
- Separate views: **MobileView** / **TabletView**
- Shared widgets and logic extracted to a common base
- No duplication of common UI parts



- **Flat Hub:** Persistent **Bottom Tab Bar** and side menu provide instant access to core modules.

- **Deep Navigation:** Sequential views (e.g., *Register*) are managed via a nested **Navigator Stack**.

